

Gradient Boosting for Spatial Panel Models with Random and Fixed Effects

Michael Balzer^{1*} and Adhen Benlahlou¹

¹Bielefeld University, Center for Mathematical Economics,
Universitätsstraße 25, Bielefeld, 33615, NW, Germany.

*Corresponding author(s). E-mail(s): michael.balzer@uni-bielefeld.de;
Contributing authors: adhen.benlahlou@uni-bielefeld.de;

Abstract

Due to the increase in data availability in urban and regional studies, various spatial panel models have emerged to model spatial panel data, which exhibit spatial patterns and spatial dependencies between observations across time. Although estimation is usually based on maximum likelihood or generalized method of moments, these methods may fail to yield unique solutions if researchers are faced with high-dimensional settings. This article proposes a model-based gradient boosting algorithm, which enables estimation with interpretable results that is feasible in low- and high-dimensional settings. Due to its modular nature, the flexible model-based gradient boosting algorithm is suitable for a variety of spatial panel models, which can include random and fixed effects. The general framework also enables data-driven model and variable selection as well as implicit regularization where the bias-variance trade-off is controlled for, thereby enhancing accuracy of prediction on out-of-sample spatial panel data. Monte Carlo experiments concerned with the performance of estimation and variable selection confirm proper functionality in low- and high-dimensional settings while real-world applications including non-life insurance in Italian districts, rice production in Indonesian farms and life expectancy in German districts illustrate the potential application.

Keywords: Spatial panel data, Error components, Gradient boosting, Statistical learning, Variable selection, Regularization

1 Introduction

Economic outcomes frequently exhibit spatial and temporal dependencies, so that changes in one geographical location can influence nearby locations and evolve over time. Linear panel data models usually ignore this so-called spatial autocorrelation, which can lead to biased estimates and invalid inference. To account for spatial autocorrelation in data, linear panel data models have been extended to spatial panel data models, explicitly incorporating the spatio-temporal dependence structure. One common approach assumes that spatial autocorrelation enters the model only through the error term, allowing for spatially autocorrelated location-specific random effects and spatially autocorrelated remainder disturbances. Such a model is referred to as the generalized spatial panel model with error components (GSPECM) (Baltagi, 2008; Baltagi et al, 2013).

Estimation of the GSPECM is typically conducted using maximum likelihood (ML) or generalized method of moments (GMM). These methods are well behaved in low-dimensional settings where the number of variables is small compared to the number of spatial units. However, in many modern applications, the number of potential variables is large while the number of spatial units grows slowly or remains fixed. In such high-dimensional settings, conventional ML and GMM estimators can be ill-defined or fail to yield unique solutions, making model and variable selection essential. Since standard ML and GMM estimators do not provide an inherent variable selection mechanisms, recent research has extended penalized estimation techniques to spatial panel models, including penalized quasi-maximum likelihood and the least absolute shrinkage and selection operator (Xia et al, 2023; Liu, 2024).

Alternatively, model-based gradient boosting has been extended to classical linear panel models (Hao et al, 2024). This iterative algorithm estimates the model by sequentially fitting base-learners to the negative gradient of a prespecified loss function, where each base-learner represents the effect of a variable and the loss function corresponds to the statistical model of interest. At each iteration, only the best-performing base-learner is selected and a small portion of its contribution is added to the current linear predictor. Early stopping provides a data-driven approach to model and variable selection, producing interpretable results at every iteration. Because of its modular and iterative nature, model-based gradient boosting remains computationally feasible even in high-dimensional settings (Bühlmann, 2006; Bühlmann and Hothorn, 2007; Mayr et al, 2014).

Although model-based gradient boosting has been applied to cross-sectional spatial regression models with autoregressive disturbances, no prior work appears to have addressed a potential extension of model-based gradient boosting for the GSPECM. This article therefore develops model-based gradient boosting for the GSPECM, building on the algorithm proposed by Balzer (2025). The algorithm is implemented in the `mboost` package, which supports gradient boosting for generalized linear, additive, and interaction models of potentially high dimensionality in R (Bühlmann and Hothorn, 2007; Hothorn et al, 2010; Hofner et al, 2014, 2015; R Core Team, 2026). Its performance is assessed via extensive Monte Carlo experiments in both low- and high-dimensional settings, focusing on estimation accuracy and variable selection. The

approach is further illustrated through three empirical applications, which are concerned with the determinants of non-life insurance in Italian provinces, rice production on Indonesian farms and life expectancy in German districts.

The rest of the article is organized as follows. Section 2 introduces the GSPECM, the associated estimator as well as the model-based gradient boosting algorithm. Monte Carlo experiments are reported in Section 3. Finally, empirical illustrations are presented in Section 4 while Section 5 concludes the article.

2 Methodology

2.1 Spatial Panel Models with Error Components

Let $N \in \mathbb{N}$ be the number of observations and $T \in \mathbb{N}$ the number of time periods in a spatial data set. For each location $i \in \{1, \dots, N\}$ and time period $t \in \{1, \dots, T\}$, consider following GSPECM

$$\begin{aligned} \mathbf{y}_t &= \mathbf{X}_t \boldsymbol{\beta} + \mathbf{W} \mathbf{X}_t \boldsymbol{\theta} + \mathbf{v}_t \\ \mathbf{v}_t &= \mathbf{u}_1 + \mathbf{u}_{2t} \\ \mathbf{u}_1 &= \rho_1 \mathbf{W} \mathbf{u}_1 + \boldsymbol{\mu} \\ \mathbf{u}_{2t} &= \rho_2 \mathbf{W} \mathbf{u}_{2t} + \boldsymbol{\epsilon}_t \end{aligned} \tag{1}$$

where $\mathbf{y}_t$ is the $N \times 1$ vector of observations, $\mathbf{X}_t$ is the $N \times P$ non-stochastic design matrix of $P \in \mathbb{N}$ exogenous variables, $\boldsymbol{\beta}$ is the time-invariant $P \times 1$ coefficient vector and $\mathbf{v}_t$ is the $N \times 1$ vector of disturbances at time t . The spatial autocorrelation enters the model in two ways in which the $N \times N$ spatial weight matrix $\mathbf{W}$ that captures spatial connections between observations plays a crucial role. A $N \times P$ non-stochastic design matrix of spatial lags of exogenous variables $\mathbf{W} \mathbf{X}_t$ and corresponding time-invariant $P \times 1$ coefficient vector $\boldsymbol{\theta}$ capture spatial autocorrelation in the explanatory variables while the disturbance term $\mathbf{v}_t$ follows an error components model with spatial autocorrelation parameters.¹ Particularly, $\mathbf{u}_1$ is the $N \times 1$ time-invariant location-specific vector of disturbances, which includes the $N \times 1$ vector of location-specific random effects $\boldsymbol{\mu}$ and $\mathbf{u}_{2t}$ is the $N \times 1$ time-varying vector of remainder disturbances, which includes the $N \times 1$ vector of idiosyncratic random innovations $\boldsymbol{\epsilon}_t$. Elements of $\boldsymbol{\mu}$ and $\boldsymbol{\epsilon}$ are independently and identically distributed according to $\boldsymbol{\mu} \sim \mathcal{N}(0, \sigma_{\boldsymbol{\mu}}^2)$ across i and $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma_{\boldsymbol{\epsilon}}^2)$ across i and t , respectively. Additionally, both disturbances terms are allowed to depend on the spatial weight matrix $\mathbf{W}$ and spatial autocorrelation parameters ρ_1 and ρ_2 , which capture spatial autocorrelation in the disturbance term.

¹Note that a spatial dependence structure in data can also be accounted for by additionally incorporating higher-order spatial lags of exogenous variables $\mathbf{W} \mathbf{W} \mathbf{X}_t$ or even $\mathbf{W} \mathbf{W} \mathbf{W} \mathbf{X}_t$ as regressors.

Stacking the cross-sections together produces the GSPECM in matrix form

$$\begin{aligned} \mathbf{y} &= \mathbf{X}\boldsymbol{\beta} + (\mathbf{I}_T \otimes \mathbf{W})\mathbf{X}\boldsymbol{\theta} + \mathbf{v} \\ \mathbf{v} &= \mathbf{L}_\mu \mathbf{u}_1 + \mathbf{u}_2 \\ \mathbf{u}_1 &= \rho_1 \mathbf{W} \mathbf{u}_1 + \boldsymbol{\mu} \\ \mathbf{u}_2 &= \rho_2 (\mathbf{I}_T \otimes \mathbf{W}) \mathbf{u}_2 + \boldsymbol{\epsilon} \end{aligned} \tag{2}$$

where $\mathbf{y} = (\mathbf{y}'_1, \dots, \mathbf{y}'_T)'$, $\mathbf{X} = [\mathbf{X}'_1, \dots, \mathbf{X}'_T]'$, $\mathbf{v} = (\mathbf{v}'_1, \dots, \mathbf{v}'_T)'$, $\mathbf{u}_2 = (\mathbf{u}'_{21}, \dots, \mathbf{u}'_{2T})'$ and $\boldsymbol{\epsilon} = (\boldsymbol{\epsilon}'_1, \dots, \boldsymbol{\epsilon}'_T)'$. Consequently, $\mathbf{I}_T$ is the $T \times T$ identity matrix while the time-invariant location-specific disturbances $\mathbf{u}_1$ are repeated in all times periods via the $NT \times N$ selector matrix $\mathbf{L}_\mu = \boldsymbol{\iota}_T \otimes \mathbf{I}_N$, in which $\boldsymbol{\iota}_T$ is the $T \times 1$ vector of ones and $\mathbf{I}_N$ is the $N \times N$ identity matrix. Proper estimation of the spatial autocorrelation parameters, the coefficients of the exogenous variables, and their corresponding spatial lags is ensured by imposing the following regularity conditions.

Assumption 1.

- (a) For all $N \geq 1$, $1 \leq i \leq N$, and $1 \leq t \leq T$, the idiosyncratic random innovations ϵ_{it} are identically and independently distributed with mean zero and variance σ_ϵ^2 where $0 < \sigma_\epsilon^2 < b_\epsilon < \infty$ and possess finite fourth moments.
- (b) For all $N \geq 1$ and $1 \leq i \leq N$, the location-specific effects μ_i are identically and independently distributed with mean zero and variance σ_μ^2 where $0 < \sigma_\mu^2 < b_\mu < \infty$ and possess finite fourth moments.
- (c) The collections of random variables $\boldsymbol{\mu}$ and $\boldsymbol{\epsilon}$ are mutually independent.

Assumption 2.

- (a) The spatial weight matrix $\mathbf{W}$ has no self-loops such that the diagonal entries satisfy $w_{ii} = 0$. The off-diagonal entries satisfy $w_{ij} = O(\frac{1}{H})$ where $\frac{H}{N} \rightarrow 0$.
- (b) For any $|\rho_1| < 1$ and $|\rho_2| < 1$, the matrices $\mathbf{I}_N - \rho_1 \mathbf{W}$ and $\mathbf{I}_N - \rho_2 \mathbf{W}$ exist and are non-singular.
- (c) The row and column sums of $\mathbf{W}$ and $(\mathbf{I}_N - \rho_1 \mathbf{W})^{-1}$ and $(\mathbf{I}_N - \rho_2 \mathbf{W})^{-1}$ are uniformly bounded in absolute value.

The GSPECM thus generalizes a variety of spatial panel models discussed in the literature. If $\rho_1 = \rho_2$, the GSPECM reduces to the spatial random effects model of [Kapoor et al \(2007\)](#) (KKP), whereas setting $\rho_1 = 0$ yields the spatial random effects model of [Anselin \(1988\)](#) (ANS). Both models are widely employed in applied work. If the spatial autocorrelation parameters are eliminated entirely $\rho_1 = \rho_2 = 0$, the GSPECM simplifies to the random effects panel data model ([Baltagi, 2008](#); [Mutl and Pfaffermayr, 2011](#); [Baltagi et al, 2013](#); [Baltagi and Liu, 2016](#)).

2.1.1 Random and Fixed Effects Specification

Combining the exogenous variables with their spatial lags into the design matrix $\mathbf{Z} = [\mathbf{X}, (\mathbf{I}_T \otimes \mathbf{W})\mathbf{X}]$ and the corresponding coefficients into $\boldsymbol{\delta} = (\boldsymbol{\beta}', \boldsymbol{\theta}')$ yields the linear predictor $\boldsymbol{\eta} = \mathbf{Z}\boldsymbol{\delta}$ and thereby the GSPECM in compact form as

$$\mathbf{y} = \boldsymbol{\eta} + \mathbf{v}, \quad \mathbb{E}(\mathbf{v}) = \mathbf{0}, \quad \text{Var}(\mathbf{v}) = \boldsymbol{\Omega}. \tag{3}$$

The following regularity condition is imposed on $\mathbf{Z}$.

Assumption 3. *The matrix $\mathbf{Z}$ has full column rank. Specifically, the limit $\lim_{NT \rightarrow \infty} \mathbf{Z}'\mathbf{Z}$ exists, is non-singular and the elements of $\mathbf{Z}$ are uniformly bounded in absolute value.*

The loss function associated with Equation (3) is the squared Mahalanobis distance of the disturbance term, arising from the generalized least squares objective function

$$f(\mathbf{y}, \boldsymbol{\eta}, \boldsymbol{\Omega}) = (\mathbf{y} - \boldsymbol{\eta})' \boldsymbol{\Omega}^{-1} (\mathbf{y} - \boldsymbol{\eta}) \quad (4)$$

such that the negative gradient of the loss function with respect to the linear predictor $\boldsymbol{\eta}$ is

$$-\frac{\partial}{\partial \boldsymbol{\eta}} f(\mathbf{y}, \boldsymbol{\eta}, \boldsymbol{\Omega}) = 2\boldsymbol{\Omega}^{-1}(\mathbf{y} - \boldsymbol{\eta}). \quad (5)$$

Let $\mathbf{A} = (\mathbf{I}_N - \rho_1 \mathbf{W})$ and $\mathbf{B} = (\mathbf{I}_N - \rho_2 \mathbf{W})$, then the distributions of the disturbance terms are given by

$$\mathbf{u}_1 = \mathbf{A}^{-1} \boldsymbol{\mu} \sim \mathcal{N}(0, \sigma_{\boldsymbol{\mu}}^2 (\mathbf{A}' \mathbf{A})^{-1}) \quad (6)$$

$$\mathbf{u}_2 = (\mathbf{I}_T \otimes \mathbf{B}^{-1}) \boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma_{\boldsymbol{\epsilon}}^2 (\mathbf{I}_T \otimes (\mathbf{B}' \mathbf{B})^{-1})). \quad (7)$$

Let $\mathbf{E}_T = \mathbf{I}_T - \bar{\mathbf{J}}_T$ and $\bar{\mathbf{J}}_T = \frac{\mathbf{J}_T}{T}$, where $\bar{\mathbf{J}}_T$ is the $T \times T$ averaging matrix and $\mathbf{J}_T$ is the $T \times T$ matrix of ones. Thus, the concentration matrix of the GSPECM is given by

$$\boldsymbol{\Omega}^{-1} = \bar{\mathbf{J}}_T \otimes [T\sigma_{\boldsymbol{\mu}}^2 (\mathbf{A}' \mathbf{A})^{-1} + \sigma_{\boldsymbol{\epsilon}}^2 (\mathbf{B}' \mathbf{B})^{-1}]^{-1} + \frac{1}{\sigma_{\boldsymbol{\epsilon}}^2} [\mathbf{E}_T \otimes (\mathbf{B}' \mathbf{B})], \quad (8)$$

yielding all necessary ingredients for the model-based gradient boosting algorithm in the random effects specification.

The random effects specification relies on the critical assumption that the unobserved time-invariant location-specific disturbances $\mathbf{u}_1$ are uncorrelated with the entries of $\mathbf{Z}$. If the assumption is violated, then $\mathbb{E}(u_{1i} | Z_{it}) \neq 0$ such that model-based gradient boosting with the random effects specification is inappropriate. Then, the fixed effects specification can be utilized instead, which has an identical representation for the GSPECM as well as the ANS and KKP random effects model (Lee and Yu, 2010). Specifically, let $\mathbf{Q} = \mathbf{E}_T \otimes \mathbf{I}_N$ denote the within transformation. This can be combined with a Cochrane–Orcutt–type spatial transformation using $(\mathbf{I}_T \otimes \mathbf{B})$, such that $\mathbf{Q}(\mathbf{I}_T \otimes \mathbf{B}) = (\mathbf{E}_T \otimes \mathbf{B})$ to obtain

$$(\mathbf{E}_T \otimes \mathbf{B})\mathbf{y} = (\mathbf{E}_T \otimes \mathbf{B})\boldsymbol{\eta} + \mathbf{Q}\boldsymbol{\epsilon} \quad \mathbb{E}(\boldsymbol{\epsilon}) = 0, \quad \text{Var}(\boldsymbol{\epsilon}) = \sigma_{\boldsymbol{\epsilon}}^2 \mathbf{I}_{NT}. \quad (9)$$

The loss function associated with Equation (9) is the squared Mahalanobis distance of the disturbance term, arising from the generalized least squares objective function

$$f(\mathbf{y}, \boldsymbol{\eta}, \boldsymbol{\Psi}) = (\mathbf{y} - \boldsymbol{\eta})' \boldsymbol{\Psi}^{-1} (\mathbf{y} - \boldsymbol{\eta}) \quad (10)$$

such that the negative gradient of the loss function with respect to the linear predictor $\boldsymbol{\eta}$ is

$$-\frac{\partial}{\partial \boldsymbol{\eta}} f(\mathbf{y}, \boldsymbol{\eta}, \boldsymbol{\Psi}) = 2\boldsymbol{\Psi}^{-1}(\mathbf{y} - \boldsymbol{\eta}). \quad (11)$$

In this case, the variance-covariance matrix of the idiosyncratic random innovations $\boldsymbol{\epsilon}$ is adjusted after inverting by the within and Cochrane–Orcutt type spatial transformation and the concentration matrix is thus given by

$$\boldsymbol{\Psi}^{-1} = \frac{1}{\sigma_{\epsilon}^2} (\mathbf{E}_T \otimes (\mathbf{B}'\mathbf{B})), \quad (12)$$

yielding all necessary ingredients for the model-based gradient boosting algorithm in the fixed effects specification (Baltagi and Liu, 2016; Balzer, 2025).²

2.1.2 Generalized Method of Moments Estimation

The feasibility of model-based gradient boosting relies on the assumption that the spatial autocorrelation parameters ρ_1 , ρ_2 and the variances of the location-specific random effects σ_{μ}^2 as well as the idiosyncratic random innovations σ_{ϵ}^2 are simultaneously apriori known. However, in real-world application settings, ρ_1 , ρ_2 , σ_{μ}^2 and σ_{ϵ}^2 are unknown, making the evaluation of $\boldsymbol{\Omega}$ and $\boldsymbol{\Psi}$, which appear in the squared Mahalanobis distance and the negative gradient, infeasible.

To obtain a feasible model-based gradient boosting for the GSPECM, ρ_1 , ρ_2 , σ_{μ}^2 and σ_{ϵ}^2 have to be replaced by corresponding estimates $\hat{\rho}_1$, $\hat{\rho}_2$, $\hat{\sigma}_{\mu}^2$ and $\hat{\sigma}_{\epsilon}^2$, which can be conveniently computed via GMM estimators. More specifically, define $\tilde{\mathbf{v}} = \mathbf{y} - \mathbf{Z}\tilde{\boldsymbol{\delta}}$ as the predictors of $\mathbf{v}$ based on a consistent estimator $\tilde{\boldsymbol{\delta}}$, ignoring random effects and spatial autocorrelation. Furthermore, define $\bar{\mathbf{u}}_2 = (\mathbf{I}_T \otimes \mathbf{W})\mathbf{u}_2$ and $\bar{\bar{\mathbf{u}}}_2 = (\mathbf{I}_T \otimes \mathbf{W})\bar{\mathbf{u}}_2$ such that $\boldsymbol{\epsilon} = \mathbf{u}_2 - \rho_2\bar{\mathbf{u}}_2$ and $\bar{\boldsymbol{\epsilon}} = \bar{\mathbf{u}}_2 - \rho_2\bar{\bar{\mathbf{u}}}_2$, then GMM estimators of ρ_2 and σ_{ϵ}^2 are based on following three moment conditions

$$\frac{1}{N(T-1)} \mathbb{E}(\boldsymbol{\epsilon}'\mathbf{Q}\boldsymbol{\epsilon}) = \sigma_{\epsilon}^2, \quad (13)$$

$$\frac{1}{N(T-1)} \mathbb{E}(\bar{\boldsymbol{\epsilon}}'\mathbf{Q}\bar{\boldsymbol{\epsilon}}) = \sigma_{\epsilon}^2 \frac{1}{N} \text{tr}(\mathbf{W}'\mathbf{W}), \quad (14)$$

$$\frac{1}{N(T-1)} \mathbb{E}(\bar{\boldsymbol{\epsilon}}'\mathbf{Q}\boldsymbol{\epsilon}) = 0. \quad (15)$$

By construction, $\mathbf{Q}(\iota_T \otimes \mathbf{u}_1) = 0$ and thus it follows that $\mathbf{v}'\mathbf{Q}\mathbf{v} = \mathbf{u}_2'\mathbf{Q}\mathbf{u}_2$, $\bar{\mathbf{v}}'\mathbf{Q}\bar{\mathbf{v}} = \bar{\mathbf{u}}_2'\mathbf{Q}\bar{\mathbf{u}}_2$ and $\bar{\bar{\mathbf{v}}}'\mathbf{Q}\bar{\bar{\mathbf{v}}} = \bar{\bar{\mathbf{u}}}_2'\mathbf{Q}\bar{\bar{\mathbf{u}}}_2$ such that a system of three equations can be obtained as

$$\boldsymbol{\Gamma}_0[\rho_2, \rho_2^2, \sigma_{\epsilon}^2]' - \boldsymbol{\gamma}_0 = 0. \quad (16)$$

²Note that the variance-covariance matrix before inverting is also adjusted by the within and Cochrane–Orcutt type spatial transformation and given by $\boldsymbol{\Psi} = \sigma_{\epsilon}^2(\mathbf{E}_T \otimes (\mathbf{B}'\mathbf{B}))$.

The expressions for $\mathbf{\Gamma}_0$ and $\boldsymbol{\gamma}_0$ are given by

$$\mathbf{\Gamma}_0 = \begin{bmatrix} \frac{2}{N(T-1)} \mathbb{E}(\bar{\mathbf{v}}' \mathbf{Q} \mathbf{v}) & -\frac{1}{N(T-1)} \mathbb{E}(\bar{\mathbf{v}}' \mathbf{Q} \bar{\mathbf{v}}) & 1 \\ \frac{2}{N(T-1)} \mathbb{E}(\bar{\mathbf{v}}' \mathbf{Q} \bar{\mathbf{v}}) & -\frac{1}{N(T-1)} \mathbb{E}(\bar{\mathbf{v}}' \mathbf{Q} \bar{\mathbf{v}}) & \frac{1}{N} \text{tr}(\mathbf{W}' \mathbf{W}) \\ \frac{1}{N(T-1)} \mathbb{E}(\bar{\mathbf{v}}' \mathbf{Q} \mathbf{v} + \bar{\mathbf{v}}' \mathbf{Q} \bar{\mathbf{v}}) & -\frac{1}{N(T-1)} \mathbb{E}(\bar{\mathbf{v}}' \mathbf{Q} \bar{\mathbf{v}}) & 0 \end{bmatrix}$$

$$\boldsymbol{\gamma}_0 = \begin{bmatrix} \frac{1}{N(T-1)} \mathbb{E}(\mathbf{v}' \mathbf{Q} \mathbf{v}) \\ \frac{1}{N(T-1)} \mathbb{E}(\bar{\mathbf{v}}' \mathbf{Q} \bar{\mathbf{v}}) \\ \frac{1}{N(T-1)} \mathbb{E}(\bar{\mathbf{v}}' \mathbf{Q} \mathbf{v}) \end{bmatrix}.$$

Replacing theoretical moment matrix and vector in Equation (16) by the corresponding sample moment matrix and vector yields

$$\mathbf{G}_0[\rho_2, \rho_2^2, \sigma_\epsilon^2]' - \mathbf{g}_0 = \boldsymbol{\xi}_0(\rho_2, \sigma_\epsilon^2). \quad (17)$$

Define $\tilde{\mathbf{v}} = (\mathbf{I}_T \otimes \mathbf{W}) \bar{\mathbf{v}}$ and $\tilde{\tilde{\mathbf{v}}} = (\mathbf{I}_T \otimes \mathbf{W}) \tilde{\mathbf{v}}$, then expressions for sample moment matrix $\mathbf{G}_0$ and vector $\mathbf{g}_0$ are given by

$$\mathbf{G}_0 = \begin{bmatrix} \frac{2}{N(T-1)} \tilde{\mathbf{v}}' \mathbf{Q} \tilde{\mathbf{v}} & -\frac{1}{N(T-1)} \tilde{\mathbf{v}}' \mathbf{Q} \tilde{\tilde{\mathbf{v}}} & 1 \\ \frac{2}{N(T-1)} \tilde{\tilde{\mathbf{v}}} \mathbf{Q} \tilde{\mathbf{v}} & -\frac{1}{N(T-1)} \tilde{\tilde{\mathbf{v}}} \mathbf{Q} \tilde{\tilde{\mathbf{v}}} & \frac{1}{N} \text{tr}(\mathbf{W}' \mathbf{W}) \\ \frac{1}{N(T-1)} (\tilde{\tilde{\mathbf{v}}} \mathbf{Q} \tilde{\mathbf{v}} + \tilde{\mathbf{v}}' \mathbf{Q} \tilde{\tilde{\mathbf{v}}}) & -\frac{1}{N(T-1)} \tilde{\tilde{\mathbf{v}}} \mathbf{Q} \tilde{\tilde{\mathbf{v}}} & 0 \end{bmatrix}$$

$$\mathbf{g}_0 = \begin{bmatrix} \frac{1}{N(T-1)} \tilde{\mathbf{v}}' \mathbf{Q} \tilde{\mathbf{v}} \\ \frac{1}{N(T-1)} \tilde{\tilde{\mathbf{v}}} \mathbf{Q} \tilde{\mathbf{v}} \\ \frac{1}{N(T-1)} \tilde{\tilde{\mathbf{v}}} \mathbf{Q} \tilde{\tilde{\mathbf{v}}} \end{bmatrix}.$$

Similarly, define $\bar{\boldsymbol{\mu}} = \mathbf{W} \boldsymbol{\mu}$, $\bar{\mathbf{u}}_1 = \mathbf{W} \mathbf{u}_1$ and $\bar{\bar{\mathbf{u}}}_1 = \mathbf{W} \bar{\mathbf{u}}_1$, then GMM estimators of ρ_1 and σ_μ^2 are based on following three moment conditions

$$\frac{1}{N} \mathbb{E}(\boldsymbol{\mu}' \boldsymbol{\mu}) = \sigma_\mu^2, \quad (18)$$

$$\frac{1}{N} \mathbb{E}(\bar{\boldsymbol{\mu}}' \bar{\boldsymbol{\mu}}) = \sigma_\mu^2 \frac{1}{N} \text{tr}(\mathbf{W}' \mathbf{W}), \quad (19)$$

$$\frac{1}{N} \mathbb{E}(\bar{\boldsymbol{\mu}}' \boldsymbol{\mu}) = 0 \quad (20)$$

such that a system of three equations can be obtained as

$$\mathbf{\Gamma}_1[\rho_1, \rho_1^2, \sigma_\mu^2]' - \boldsymbol{\gamma}_1 = 0. \quad (21)$$

Conveniently, theoretical moment matrix $\mathbf{\Gamma}_1$ and vector $\boldsymbol{\gamma}_1$ can be expressed in terms of $\mathbf{v}$, $\bar{\mathbf{v}}$ and $\tilde{\mathbf{v}}$ by utilizing $\mathbf{S} = \left(\bar{\mathbf{J}}_T - \frac{1}{T-1} \mathbf{E}_T \right) \otimes \mathbf{I}_N$ as

$$\mathbf{\Gamma}_1 = \begin{bmatrix} \frac{2}{NT} \mathbb{E}(\bar{\mathbf{v}}' \mathbf{S} \mathbf{v}) & -\frac{1}{NT} \mathbb{E}(\bar{\mathbf{v}}' \mathbf{S} \bar{\mathbf{v}}) & 1 \\ \frac{2}{NT} \mathbb{E}(\tilde{\mathbf{v}}' \mathbf{S} \bar{\mathbf{v}}) & -\frac{1}{NT} \mathbb{E}(\tilde{\mathbf{v}}' \mathbf{S} \tilde{\mathbf{v}}) & \frac{1}{N} \text{tr}(\mathbf{W}' \mathbf{W}) \\ \frac{1}{NT} \mathbb{E}(\tilde{\mathbf{v}}' \mathbf{S} \mathbf{v} + \bar{\mathbf{v}}' \mathbf{S} \bar{\mathbf{v}}) & -\frac{1}{NT} \mathbb{E}(\tilde{\mathbf{v}}' \mathbf{S} \bar{\mathbf{v}}) & 0 \end{bmatrix}$$

$$\boldsymbol{\gamma}_1 = \begin{bmatrix} \frac{1}{NT} \mathbb{E}(\mathbf{v}' \mathbf{S} \mathbf{v}) \\ \frac{1}{NT} \mathbb{E}(\bar{\mathbf{v}}' \mathbf{S} \bar{\mathbf{v}}) \\ \frac{1}{NT} \mathbb{E}(\bar{\mathbf{v}}' \mathbf{S} \mathbf{v}) \end{bmatrix}.$$

Replacing theoretical moment matrix and vector in Equation (21) by the corresponding sample moment matrix and vector yields

$$\mathbf{G}_1[\rho_1, \rho_1^2, \sigma_\mu^2]' - \mathbf{g}_1 = \boldsymbol{\xi}_1(\rho_1, \sigma_\mu^2). \quad (22)$$

where expressions for sample moment matrix $\mathbf{G}_1$ and vector $\mathbf{g}_1$ are given by

$$\mathbf{G}_1 = \begin{bmatrix} \frac{2}{NT} \tilde{\mathbf{v}}' \mathbf{S} \tilde{\mathbf{v}} & -\frac{1}{NT} \tilde{\mathbf{v}}' \mathbf{S} \tilde{\mathbf{v}} & 1 \\ \frac{2}{NT} \tilde{\mathbf{v}}' \mathbf{S} \bar{\mathbf{v}} & -\frac{1}{NT} \tilde{\mathbf{v}}' \mathbf{S} \tilde{\mathbf{v}} & \frac{1}{N} \text{tr}(\mathbf{W}' \mathbf{W}) \\ \frac{1}{NT} (\tilde{\mathbf{v}}' \mathbf{S} \bar{\mathbf{v}} + \bar{\mathbf{v}}' \mathbf{S} \tilde{\mathbf{v}}) & -\frac{1}{NT} \tilde{\mathbf{v}}' \mathbf{S} \tilde{\mathbf{v}} & 0 \end{bmatrix}$$

$$\mathbf{g}_1 = \begin{bmatrix} \frac{1}{NT} \tilde{\mathbf{v}}' \mathbf{S} \tilde{\mathbf{v}} \\ \frac{1}{NT} \tilde{\mathbf{v}}' \mathbf{S} \bar{\mathbf{v}} \\ \frac{1}{NT} \tilde{\mathbf{v}}' \mathbf{S} \bar{\mathbf{v}} \end{bmatrix}.$$

The GMM estimators can thus be obtained by solving following optimization problems via non-linear least squares estimation

$$(\hat{\rho}_2, \hat{\sigma}_\epsilon^2) = \arg \min_{\rho_2, \sigma_\epsilon^2} \boldsymbol{\xi}_0(\rho_2, \sigma_\epsilon^2)' \boldsymbol{\xi}_0(\rho_2, \sigma_\epsilon^2) \quad (23)$$

$$(\hat{\rho}_1, \hat{\sigma}_\mu^2) = \arg \min_{\rho_1, \sigma_\mu^2} \boldsymbol{\xi}_1(\rho_1, \sigma_\mu^2)' \boldsymbol{\xi}_1(\rho_1, \sigma_\mu^2). \quad (24)$$

Let Assumptions 1 to 4 hold.

Assumption 4. *The smallest eigenvalues of $\mathbf{\Gamma}'_0 \mathbf{\Gamma}_0$ and $\mathbf{\Gamma}'_1 \mathbf{\Gamma}_1$ are bounded away from zero.*

Then, the non-linear least squares estimators $\hat{\rho}_1$, $\hat{\rho}_2$, $\hat{\sigma}_\mu^2$ and $\hat{\sigma}_\epsilon^2$ are consistent estimators of ρ_1 , ρ_2 , σ_μ^2 and σ_ϵ^2 in the sense that $(\hat{\rho}_2, \hat{\sigma}_\epsilon^2) \rightarrow^p (\rho_2, \sigma_\epsilon^2)$ and $(\hat{\rho}_1, \hat{\sigma}_\mu^2) \rightarrow^p (\rho_1, \sigma_\mu^2)$ for $N \rightarrow \infty$ sufficiently large (Kapoor et al, 2007; Baltagi and Liu, 2016).

2.2 Feasible Model-based Gradient Boosting

2.2.1 Algorithm and Implementation

Given estimates of the spatial autocorrelation parameters and the variances, the empirical counterparts $\hat{\Omega}$ and $\hat{\Psi}$ of Ω and Ψ can be computed, enabling a feasible implementation of the model-based gradient boosting algorithm, where the loss function and its corresponding negative gradient determine the updates of the linear predictor. In particular, model-based gradient boosting initializes with an empty model, computes the negative gradient vector of the loss function at the current estimate of the linear predictor and then continues by sequentially fitting specified base-learners to the computed negative gradient vector. In this case, the each base-learner describes the functional form of a single variable occurring in the design matrix, which can be identified as a simple linear model for the GSPECM. For each specified base-learner, the residual sum of squares is computed and the one which minimizes the residual sum of squares is selected. Afterwards, an update in the linear predictor is performed by adding a small fraction of the best-performing base-learner. This algorithm is then continued iteratively until the number of boosting iterations m_{stop} is reached (Friedman, 2001; Bühlmann and Hothorn, 2007; Mayr et al, 2014; Hepp et al, 2016).

To implement model-based gradient boosting for cross-sectional spatial regression models, a so-called spatial error family can be utilized, which incorporates the concentration matrix into the updates of the algorithm. Although the proposed spatial error family is quite general and can be utilized for model-based gradient boosting for the GSPECM without any alteration, the implementation has various limitations. First, performance of variable selection and regularization is influenced by magnitude of the spatial autocorrelation parameters through the concentration matrix in the updates. Particularly, large positive values of ρ_1 and ρ_2 may lead to stronger regularization and more pronounced variable selection, which can have a negative impact on the bias-variance trade-off. Second, the concentration matrix severely destabilizes the negative gradient vector in the updates. Although this is rarely a concern in the case of cross-sectional spatial regression models, the increase in complexity in the structure of the concentration matrix for the GSPECM leads to situations, in which the model-based gradient boosting takes very tiny steps towards the solution, thereby requiring a large number of number of boosting iterations. Third, in practice, the implementation via the spatial error family is slow and computationally expensive. Fourth, a formal theoretical justification of model-based gradient boosting with the squared Mahalanobis distance, where the concentration matrix is induced by spatial autocorrelation is still missing (Balzer, 2025).

To circumvent the discussed limitations, a Cochrane–Orcutt type spatial transformation can be directly applied to the spatial data such that the concentration matrix is not required within the iterations of the model-based gradient boosting algorithm. Consequently, the squared Mahalanobis distance reduces to the L_2 loss function and the negative gradient vector to the negative residuals of the Cochrane–Orcutt

type spatial transformed variables.³ The model-based gradient boosting algorithm thus gains in computational efficiency and formal theoretical justification in low- and high-dimensional settings (see, for example, Zhang and Yu (2005); Bühlmann (2006); Bühlmann and Hothorn (2007); Hao et al (2024)).⁴ A summary of the Cochrane–Orcutt type spatial transformed variables as well as the corresponding ingredients for model-based gradient boosting in the random and fixed specification can be seen in Table 1.

Table 1 Response variable, design matrix, linear predictor, loss function, negative gradient and concentration matrix utilized in the model-based gradient boosting algorithm for the generalized spatial panel model with error components (GSPECM).

Random effects specification		
	Original	Transformed
Response variable	$\mathbf{y}$	$\mathbf{y}_* = \hat{\Omega}^{-\frac{1}{2}} \mathbf{y}$
Design matrix	$\mathbf{Z}$	$\mathbf{Z}_* = \hat{\Omega}^{-\frac{1}{2}} \mathbf{Z}$
Linear predictor	$\boldsymbol{\eta} = \mathbf{Z}\boldsymbol{\delta}$	$\boldsymbol{\eta}_* = \mathbf{Z}_*\boldsymbol{\delta}$
Loss function	$f(\mathbf{y}, \boldsymbol{\eta}, \hat{\Omega}) = (\mathbf{y} - \boldsymbol{\eta})' \hat{\Omega}^{-1} (\mathbf{y} - \boldsymbol{\eta})$	$f(\mathbf{y}_*, \boldsymbol{\eta}_*) = (\mathbf{y}_* - \boldsymbol{\eta}_*)' (\mathbf{y}_* - \boldsymbol{\eta}_*)$
Negative gradient	$-\frac{\partial}{\partial \boldsymbol{\eta}} f(\mathbf{y}, \boldsymbol{\eta}, \hat{\Omega}) = 2\hat{\Omega}^{-1} (\mathbf{y} - \boldsymbol{\eta})$	$-\frac{\partial}{\partial \boldsymbol{\eta}_*} f(\mathbf{y}_*, \boldsymbol{\eta}_*) = 2(\mathbf{y}_* - \boldsymbol{\eta}_*)$
Concentration matrix	$\hat{\Omega}^{-1} = \bar{\mathbf{J}}_T \otimes [T\hat{\sigma}_\mu^2(\hat{\mathbf{A}}'\hat{\mathbf{A}})^{-1} + \hat{\sigma}_\epsilon^2(\hat{\mathbf{B}}'\hat{\mathbf{B}})^{-1}]^{-1} + \frac{1}{\hat{\sigma}_\epsilon^2} [\mathbf{E}_T \otimes (\hat{\mathbf{B}}'\hat{\mathbf{B}})]$ with $\hat{\mathbf{A}} = (\mathbf{I}_N - \hat{\rho}_1 \mathbf{W})$ and $\hat{\mathbf{B}} = (\mathbf{I}_N - \hat{\rho}_2 \mathbf{W})$	
Fixed effects specification		
	Original	Transformed
Response variable	$\mathbf{y}$	$\mathbf{y}_* = (\mathbf{E}_T \otimes \hat{\mathbf{B}}) \mathbf{y}$
Design matrix	$\mathbf{Z}$	$\mathbf{Z}_* = (\mathbf{E}_T \otimes \hat{\mathbf{B}}) \mathbf{Z}$
Linear predictor	$\boldsymbol{\eta} = \mathbf{Z}\boldsymbol{\delta}$	$\boldsymbol{\eta}_* = \mathbf{Z}_*\boldsymbol{\delta}$
Loss function	$f(\mathbf{y}, \boldsymbol{\eta}, \hat{\Psi}) = (\mathbf{y} - \boldsymbol{\eta})' \hat{\Psi}^{-1} (\mathbf{y} - \boldsymbol{\eta})$	$f(\mathbf{y}_*, \boldsymbol{\eta}_*) = (\mathbf{y}_* - \boldsymbol{\eta}_*)' (\mathbf{y}_* - \boldsymbol{\eta}_*)$
Negative gradient	$-\frac{\partial}{\partial \boldsymbol{\eta}} f(\mathbf{y}, \boldsymbol{\eta}, \hat{\Psi}) = 2\hat{\Psi}^{-1} (\mathbf{y} - \boldsymbol{\eta})$	$-\frac{\partial}{\partial \boldsymbol{\eta}_*} f(\mathbf{y}_*, \boldsymbol{\eta}_*) = 2(\mathbf{y}_* - \boldsymbol{\eta}_*)$
Concentration matrix	$\hat{\Psi}^{-1} = \frac{1}{\hat{\sigma}_\epsilon^2} (\mathbf{E}_T \otimes (\hat{\mathbf{B}}'\hat{\mathbf{B}}))$ with $\hat{\mathbf{B}} = (\mathbf{I}_N - \hat{\rho}_2 \mathbf{W})$	

Notes: Entries are reported for the random effects specification and the fixed effects specification with the model on the untransformed scale (Original) and on the Cochrane–Orcutt type spatial transformation scale (Transformed).

Applying the Cochrane–Orcutt-type spatial transformation reduces the general model-based gradient boosting algorithm to L_2 -boosting (LTB) using the L_2 loss function, as discussed by Hao et al (2024) in the context of linear panel data models to address the many-instrument problem. The following algorithmic outline shows how LTB can be directly applied for the GSPECM via the Cochrane–Orcutt type transformation:

³Essentially, the squared Mahalanobis distance can be interpreted as the L_2 loss function weighted by the concentration matrix of the disturbance term. Consequently, the negative gradient corresponds to the weighted negative residuals.

⁴Note that such a Cochrane–Orcutt type spatial transformation can also be applied in cross-sectional spatial regression models with autoregressive disturbances, thereby ensuring computational efficiency and theoretical justification of model-based gradient boosting in low- and high-dimensional linear settings.

1. Set $m = 0$. Choose offset values for the linear predictor as $\hat{\boldsymbol{\eta}}_*^{[0]} = (\mathbf{0})_{\{i=1, \dots, NT\}}$ and specify the set of base-learners as linear models $\delta_1 \mathbf{Z}_{*1}, \dots, \delta_K \mathbf{Z}_{*K}$, where K is the cardinality of the set of base-learners.
2. For $m = 1$ to m_{stop} :
 - (a) Calculate the residuals of the L_2 loss function at the current estimate of the linear predictor $\hat{\boldsymbol{\eta}}_*^{[m-1]}$

$$\mathbf{d}^{[m]} = \left(d_i^{[m]} \right)_{i=1, \dots, NT} = \left(- \frac{\partial}{\partial \boldsymbol{\eta}_*} f(\mathbf{y}_*, \boldsymbol{\eta}_*) \bigg|_{\boldsymbol{\eta}_* = \hat{\boldsymbol{\eta}}_*^{[m-1]}} \right).$$

- (b) Fit each specified linear model base-learner to the residuals separately.
- (c) Evaluate the residual sum of squares for each base-learner and pick component j_* that fits residuals best

$$j_* = \underset{1 \leq j \leq K}{\operatorname{argmin}} \sum_{i=1}^{NT} \left(d_i^{[m]} - \hat{\delta}_j^m \mathbf{Z}_{*j} \right)^2.$$

- (d) Update the estimate of the linear predictor $\hat{\boldsymbol{\eta}}_*^{[m]}$ based on j_* via

$$\hat{\boldsymbol{\eta}}_*^{[m]} = \hat{\boldsymbol{\eta}}_*^{[m-1]} + s \cdot \hat{\delta}_{j_*}^m \mathbf{Z}_{*j_*},$$

where s is the learning rate.

2.2.2 Optimal Stopping Criterion and Post-hoc Deselection

In model-based gradient boosting, model and variable selection are controlled by the stopping criterion m_{stop} , which governs the bias-variance trade-off. Stopping too early increases bias, whereas excessive iterations inflate variance and induce overfitting. Arbitrary selection of m_{stop} compromises both predictive power and the reliability of coefficient estimates, making the identification of an optimal stopping criterion m_{opt} essential.

In cross-sectional linear models, m_{opt} is typically determined using information criteria, such as the Akaike information criterion, or resampling techniques, including cross-validation, bootstrapping, or subsampling. For grouped, longitudinal, or panel data, extensions have been proposed in the form of an adjusted Akaike information criterion and an adjusted cross-validation criterion (Hao et al, 2024; Knieper et al, 2025). However, the adjusted criteria rely on assumptions that are violated in the presence of spatial dependence structure in data. As pointed out by Schratz et al (2019), random cross-validation presumes independence of observations when constructing folds. In spatial data, observations correspond to geographic locations and are typically spatially autocorrelated, such that random fold assignment places nearby observations in both the training and test sets. This violation leads to systematically underestimated prediction errors and unreliable model and variable selection. Furthermore, the adjusted Akaike information criterion is infeasible in this context, as it requires the

computation of the degrees of freedom in each boosting iteration, which are currently available only for linear models without spatial dependence.

In contrast, spatial cross-validation techniques provide a coherent alternative for selecting m_{opt} in the presence of a spatial dependence structure in data. Instead of randomly assigning observations to folds, spatial cross-validation constructs training and test sets that are spatially separated. For example, [Brenning \(2012\)](#) generates spatially disjoint folds by clustering spatial polygons based on geodesic distances, while [Valavi et al \(2019\)](#) partition the study region into spatial blocks or buffer zones. In spatial panel models, the temporal dimension can be additionally exploited by leaving out entire time periods, or by jointly excluding spatial and temporal units ([Schratz et al, 2024](#)). By preventing spatially proximate observations from appearing in both training and test sets, spatial cross-validation avoids downward-biased prediction error estimates, which accurately reflects out-of-sample performance under spatial dependence ([Balzer, 2025](#)).

However, standard and spatial cross-validation techniques often select too many variables, resulting in less parsimonious models ([Mayr et al, 2012](#)). To obtain more parsimonious models and prevent overfitting, the deselection algorithm of [Strömer et al \(2022\)](#) is applied to LTB for the GSPECM. The algorithm is conducted in two stages. First, model-based gradient boosting is run once and m_{opt} is selected using spatial cross-validation techniques. Second, the contribution of each base-learner to the cumulative reduction in empirical risk up to m_{opt} is quantified. Base-learners with negligible contributions are removed and LTB is rerun using only the retained variables while keeping m_{opt} fixed.

Specifically, let $\mathbb{1}(\cdot)$ denote the indicator function, $r^{[m]}$ the empirical risk based on the L_2 -loss, $(r^{[m-1]} - r^{[m]})$ the risk reduction and $j_*^{[m]}$ the base-learner selected in iteration m . The attributable risk reduction for base-learner j after m_{opt} boosting iterations is defined as

$$R_j = \sum_{m=1}^{m_{\text{opt}}} \mathbb{1}\{j = j_*^{[m]}\} (r^{[m-1]} - r^{[m]}), \quad j = 1, \dots, K. \quad (25)$$

The attributable risk reduction R_j quantifies the portion of the total empirical risk reduction attributable to base-learner j across m_{opt} . The total risk reduction is given by $(r^{[0]} - r^{[m_{\text{opt}]})$ and a base-learner j is removed if its relative contribution falls below a pre-specified threshold $\tau \in (0, 1)$

$$R_j < \tau (r^{[0]} - r^{[m_{\text{opt}]})}. \quad (26)$$

Equation (26) ensures that only base-learners contributing at least a fraction τ of the total risk reduction are retained in the model. The specific choice of τ directly controls the sparsity of the final model. Smaller values retain more variables, while larger values enforce more aggressive deselection. For linear and spatial regression models, relative small thresholds $\tau \leq 0.025$ have been found to be appropriate ([Strömer et al, 2022](#); [Balzer, 2025](#)).

3 Monte Carlo Experiments

3.1 Study Design

To assess the performance of LTB for the GSPECM, Monte Carlo experiments are conducted. In particular, both the accuracy of parameter estimation and the effectiveness of variable selection and deselection are evaluated in low- and high-dimensional settings. The number of time periods is fixed at $T = 5$, in which the number of observations is set to $N = 100$, resulting in a total of $NT = 500$ observations. The number of independent variables is varied between $K = 40$ and $K = 800$, corresponding to a low- ($NT > K$) and high-dimensional ($NT < K$) setting, respectively. Let $\boldsymbol{\iota}_{NT}$ be a $NT \times 1$ vector of ones, then the data generating process is given by

$$\begin{aligned}\mathbf{y} &= \boldsymbol{\iota}_{NT} + 3.5\mathbf{x}_1 - 2.5\mathbf{x}_2 - 4(\mathbf{I}_T \otimes \mathbf{W})\mathbf{x}_1 + 3(\mathbf{I}_T \otimes \mathbf{W})\mathbf{x}_2 + \mathbf{v} \\ \mathbf{v} &= \mathbf{L}_\mu \mathbf{u}_1 + \mathbf{u}_2 \\ \mathbf{u}_1 &= \rho_1 \mathbf{W} \mathbf{u}_1 + \boldsymbol{\mu} \\ \mathbf{u}_2 &= \rho_2 (\mathbf{I}_T \otimes \mathbf{W}) \mathbf{u}_2 + \boldsymbol{\epsilon}.\end{aligned}$$

Entries of the independent variables x_{it} are generated according to $x_{it} = \zeta_i + \kappa_{it}$ where $\zeta_i \sim \mathcal{U}(-7.5, 7.5)$ and $\kappa_{it} \sim \mathcal{U}(-5, 5)$ are independently and identically drawn from the uniform distribution. The location-specific random effects and idiosyncratic random innovations are drawn independently and identically from the normal distribution $\mu_i \sim \mathcal{N}(0, \sigma_\mu^2)$ and $\epsilon_{it} \sim \mathcal{N}(0, \sigma_\epsilon^2)$ with $\sigma_\mu^2 = \sigma_\epsilon^2 = 10$. The spatial weight matrix $\mathbf{W}$ has a 10-nearest neighbor structure where each location is connected to its ten geographically closest neighbors based on constant centroids of the spatial polygons available in the shapefile, which is based on the 100 counties in North Carolina. Afterward, $\mathbf{W}$ is row-normalized such that each row sums up to one⁵. Spatial autocorrelation parameters vary over a set of $\rho_1, \rho_2 \in \{-0.8, -0.6, -0.4, -0.2, 0, 0.2, 0.4, 0.6, 0.8\}$. In LTB, the corresponding base-learners are specified as simple linear regression models due to the nature of the data-generating process. The learning rate is set to $s = 0.1$, as is standard in the literature (see, for example, Schmid and Hothorn (2008); Mayr et al (2012); Hofner et al (2014)). The optimal stopping criterion m_{opt} is found by minimizing the empirical risk via 5-fold spatial cross-validation as proposed by Brenning (2012). The threshold parameter in deselection is set to $\tau = 0.01$. Results are reported for ML, GMM, LTB and LTB with post-hoc deselection (DES) for the random effects specification (Random) and fixed effects specification (Fixed) respectively. In each Monte Carlo experiment, a total of $N_{\text{sim}} = 100$ repetitions are conducted.

Regarding the performance of variable selection and deselection, the criteria are chosen based on the confusion matrix. In particular, the reported variable selection criteria are the true positive rate (TPR), which is the proportion of correctly selected variables out of all true informative variables and the true negative rate (TNR), which is the proportion of correctly non-selected variables out of all true non-informative variables (Stehman, 1997).

⁵The shapefile for the North Carolina counties is available in the `spData` package in R (Pebesma and Bivand, 2023; Bivand et al, 2025).

Estimation accuracy is evaluated by reporting the mean squared error (MSE) of δ defined as

$$\text{MSE} = \frac{1}{N_{\text{sim}}} \sum_{i=1}^{N_{\text{sim}}} (\hat{\delta}_i - \delta)^2$$

For all proposed performance criteria, lower values are always preferred (Morris et al, 2019).

Monte Carlo experiments are conducted in the programming language R (R Core Team, 2026). The MLE and GMM estimation of the GSPECM in the low-dimensional setting is performed via the `splm` package (Millo and Piras, 2012; Bivand et al, 2021). Model-based gradient boosting for generalized, additive and interaction models can be found in the `mboost` package (Bühlmann and Hothorn, 2007; Hothorn et al, 2010; Hofner et al, 2014, 2015). Spatial cross-validation is performed via the `sperrorest` package (Brenning, 2012). An implementation for LTB for the GSPECM incorporating the deselection algorithm and the R code for reproducibility of all Monte Carlo experiments can be found in the GitHub repository <https://github.com/micbalz/SpatPanelRegBoost>.

3.2 Results

3.2.1 Low-dimensional Linear Setting

In Table 2, the results for the performance of estimation and variable selection in the low-dimensional linear settings over $N_{\text{sim}} = 100$ repetitions are presented. Across all combinations of the spatial autocorrelation parameters (ρ_1, ρ_2) , TPR equals one for all methods in both random and fixed effects specifications, indicating that all informative variables are selected in this low-dimensional linear setting. Differences between methods arise exclusively from their ability to correctly exclude non-informative variables, as reflected by the TNR and from their estimation accuracy as measured by the MSE.

ML and GMM estimators exhibit a TNR of zero throughout, which is unsurprising since these methods do not encompass an inherent variable selection mechanism. In contrast, LTB substantially improves variable selection performance, achieving moderate to high TNR values that generally increase with the strength of spatial dependence. DES further improves on sparsity, yielding perfect exclusion of non-informative variables across all (ρ_1, ρ_2) combinations in both random and fixed effects specifications.

These improvements in variable selection translate directly into gains in estimation accuracy. While ML and GMM display comparatively large MSE values, which increase with stronger spatial autocorrelation, both LTB and DES lead to considerable reductions in MSE values. Among them, DES achieves the lowest MSE values across all (ρ_1, ρ_2) combinations. The fixed effects specification generally exhibits larger MSE values than the corresponding random effects specification.

Table 2 Performance of estimation and variable selection in the low-dimensional linear setting ($N = 100$, $T = 5$, $NT = 500$, $K = 40$) for a set of spatial autocorrelation parameters (ρ_1, ρ_2) with $N_{\text{sim}} = 100$ repetitions.

ρ_1	ρ_2		ML	Random			Fixed		
				GMM	LTB	DES	GMM	LTB	DES
-0.2	0.2	TPR	1.000	1.000	1.000	1.000	1.000	1.000	1.000
		TNR	0.000	0.000	0.718	1.000	0.000	0.764	1.000
		MSE	0.321	0.338	0.108	0.043	0.515	0.135	0.055
-0.4	0.4	TPR	1.000	1.000	1.000	1.000	1.000	1.000	1.000
		TNR	0.000	0.000	0.743	1.000	0.000	0.791	1.000
		MSE	0.355	0.374	0.107	0.045	0.563	0.142	0.062
-0.6	0.6	TPR	1.000	1.000	1.000	1.000	1.000	1.000	1.000
		TNR	0.000	0.000	0.761	1.000	0.000	0.803	1.000
		MSE	0.427	0.474	0.123	0.052	0.705	0.148	0.074
-0.8	0.8	TPR	1.000	1.000	1.000	1.000	1.000	1.000	1.000
		TNR	0.000	0.000	0.808	1.000	0.000	0.824	1.000
		MSE	0.503	0.606	0.125	0.064	0.915	0.158	0.086
0	0	TPR	1.000	1.000	1.000	1.000	1.000	1.000	1.000
		TNR	0.000	0.000	0.667	1.000	0.000	0.758	1.000
		MSE	0.308	0.341	0.125	0.038	0.429	0.136	0.059
0.2	-0.2	TPR	1.000	1.000	1.000	1.000	1.000	1.000	1.000
		TNR	0.000	0.000	0.689	1.000	0.000	0.742	1.000
		MSE	0.268	0.295	0.100	0.034	0.395	0.122	0.047
0.4	-0.4	TPR	1.000	1.000	1.000	1.000	1.000	1.000	1.000
		TNR	0.000	0.000	0.714	1.000	0.000	0.738	1.000
		MSE	0.270	0.319	0.115	0.042	0.371	0.133	0.052
0.6	-0.6	TPR	1.000	1.000	1.000	1.000	1.000	1.000	1.000
		TNR	0.000	0.000	0.723	1.000	0.000	0.725	1.000
		MSE	0.237	0.311	0.101	0.039	0.383	0.122	0.039
0.8	-0.8	TPR	1.000	1.000	1.000	1.000	1.000	1.000	1.000
		TNR	0.000	0.000	0.757	1.000	0.000	0.733	1.000
		MSE	0.243	0.378	0.127	0.049	0.416	0.137	0.054

Notes: Reported are the root mean squared error of δ (MSE), true positive rate (TPR), and true negative rate (TNR) for maximum likelihood (ML), generalized method of moments (GMM), L_2 -boosting (LTB) and L_2 -boosting with post-hoc deselection (DES) for the random effects specification (Random) and fixed effects specification (Fixed).

3.2.2 High-dimensional Linear Setting

In Table 3, the results for the performance of estimation and variable selection in the high-dimensional linear settings over $N_{\text{sim}} = 100$ repetitions are presented. Similar to the low-dimensional linear settings, both LTB and DES achieve a TPR equal to one under random and fixed effects specifications across all combinations of the spatial autocorrelation parameters (ρ_1, ρ_2) . This indicates that all informative variables are successfully identified even in the high-dimensional linear setting, highlighting proper functionality of LTB and DES.

Differences between LTB and DES are primarily reflected in their ability to exclude non-informative variables and in their estimation accuracy. ML and GMM estimators are not applicable in this setting and are therefore omitted. For LTB, the TNR is high, ranging between approximately 0.96 and 0.98, and shows a mild increase with the strength of spatial dependence. DES further improves variable selection performance,

yielding perfect exclusion of non-informative variables across all (ρ_1, ρ_2) combinations in both random and fixed effect specifications.

Table 3 Performance of estimation and variable selection in the high-dimensional linear setting ($N = 100$, $T = 5$, $NT = 500$, $K = 800$) for a set of spatial autocorrelation parameters (ρ_1, ρ_2) with $N_{\text{sim}} = 100$ repetitions.

ρ_1	ρ_2		ML	Random			Fixed		
				GMM	LTB	DES	GMM	LTB	DES
-0.2	0.2	TPR	-	-	1.000	1.000	-	1.000	1.000
		TNR	-	-	0.971	1.000	-	0.968	1.000
		MSE	-	-	0.190	0.056	-	0.262	0.067
-0.4	0.4	TPR	-	-	1.000	1.000	-	1.000	1.000
		TNR	-	-	0.974	1.000	-	0.970	1.000
		MSE	-	-	0.207	0.069	-	0.285	0.081
-0.6	0.6	TPR	-	-	1.000	1.000	-	1.000	1.000
		TNR	-	-	0.979	1.000	-	0.974	1.000
		MSE	-	-	0.182	0.066	-	0.240	0.084
-0.8	0.8	TPR	-	-	1.000	1.000	-	1.000	1.000
		TNR	-	-	0.980	1.000	-	0.978	1.000
		MSE	-	-	0.183	0.072	-	0.252	0.106
0	0	TPR	-	-	1.000	1.000	-	1.000	1.000
		TNR	-	-	0.974	1.000	-	0.970	1.000
		MSE	-	-	0.183	0.056	-	0.263	0.064
0.2	-0.2	TPR	-	-	1.000	1.000	-	1.000	1.000
		TNR	-	-	0.971	1.000	-	0.965	1.000
		MSE	-	-	0.193	0.056	-	0.284	0.063
0.4	-0.4	TPR	-	-	1.000	1.000	-	1.000	1.000
		TNR	-	-	0.971	1.000	-	0.962	1.000
		MSE	-	-	0.205	0.060	-	0.335	0.071
0.6	-0.6	TPR	-	-	1.000	1.000	-	1.000	1.000
		TNR	-	-	0.972	1.000	-	0.961	1.000
		MSE	-	-	0.214	0.069	-	0.358	0.081
0.8	-0.8	TPR	-	-	1.000	1.000	-	1.000	1.000
		TNR	-	-	0.974	1.000	-	0.960	1.000
		MSE	-	-	0.209	0.069	-	0.375	0.087

Notes: Reported are the root mean squared error of δ (MSE), true positive rate (TPR), and true negative rate (TNR) for maximum likelihood (ML), generalized method of moments (GMM), L_2 -boosting (LTB) and L_2 -boosting with post-hoc deselection (DES) for the random effects specification (Random) and fixed effects specification (Fixed).

Gains in variable selection performance come with corresponding improvements in estimation accuracy. Although LTB attains moderate MSE values, DES achieves lowest MSE values across all (ρ_1, ρ_2) combinations. Similar to the low-dimensional linear setting, MSE values are generally higher under the fixed effects specification than under the random effects specification.

Overall, the results demonstrate that in both low and high-dimensional linear settings, LTB combined with DES effectively balances perfect detection of informative variables with accurate exclusion of non-informative variables, resulting in substantial efficiency gains in terms of MSE over classical ML and GMM estimators under both random and fixed effects specifications.

4 Empirical Illustrations

4.1 Non-life Insurance in Italian Provinces

The data set on non-life insurance consumption across Italian provinces was first examined by [Millo and Carmeci \(2011\)](#) and later revisited by [Millo \(2014\)](#) and [Millo \(2022\)](#). It focuses on identifying the determinants of per capita equilibrium consumption of non-life insurance across $N_{\text{Italy}} = 103$ Italian provinces over the period 1998 – 2002, corresponding to $T_{\text{Italy}} = 5$ years. The data set contains a broad set of province-level socioeconomic characteristics, summarized in Table 4. Since most of the variables are either time-invariant or exhibit only limited variation over time, a fixed effects specification is not well suited.

Table 4 Data type and description of dependent and independent variables for non-life insurance consumption across Italian provinces data set (see, [Millo and Carmeci \(2011\)](#); [Millo and Piras \(2012\)](#); [Millo \(2014, 2022\)](#)).

Variable	Type	Description
PPCD	Numeric	Real per capita premiums in 2000 euros, non-life insurance excluding mandatory motor third-party liability
AGEN	Numeric	Density of insurance agencies per 1000 inhabitants
BANK	Numeric	Real bank deposits per capita
DEN	Numeric	Density of inhabitants per square kilometre
FAM	Numeric	Average number of family members
INEF	Numeric	Judicial inefficiency: average years to settle first degree of civil case
RGDP	Numeric	Real gross domestic product per capita
RIRS	Numeric	Real interest rate on borrowing
SCHOOL	Numeric	Share of people with second-grade schooling or more
TRUST	Numeric	Survey results to the question "do you trust others?"
VAAGR	Numeric	Share of value added, agricultural sector

Thus, to investigate the determinants of per capita equilibrium consumption of non-life insurance from a data-driven variable selection perspective via LTB and DES, a variety of random effects models, namely ANS, KKP and GSPECM are considered. For additional comparison, results for classical GMM estimation are reported. All independent variables presented in Table 4 and corresponding spatial lags of independent variables as well as an intercept are included in the estimation, leading to a low-dimensional linear setting with 21 coefficients to estimate in a non-parsimonious model. Furthermore, continuous variables are transformed via centering and scaling to ensure comparability across variables in terms of the scale and proper convergence of LTB. The setup for LTB and DES follows closely the Monte Carlo experiments in Section 3, such that the learning rate is set to $s = 0.1$ and the threshold to $\tau = 0.01$. The optimal stopping criterion m_{opt} is found by minimizing the empirical risk via 10-fold spatial cross-validation as discussed by [Brenning \(2012\)](#).⁶

⁶The shapefile of the Italian provinces is provided freely by the Italian National Institute of Statistics and can be found at <https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/>.

Table 5 Coefficient estimates across different estimation strategies for the Italian non-life insurance data set.

	GMM			LTB			DES		
	ANS	KKP	GSPECM	ANS	KKP	GSPECM	ANS	KKP	GSPECM
$\hat{\rho}_1$	0.000	0.277	0.337	0.000	0.277	0.337	0.000	0.277	0.337
$\hat{\rho}_2$	0.277	0.277	0.277	0.277	0.277	0.277	0.277	0.277	0.277
$\hat{\sigma}_\mu$	0.080	0.088	0.079	0.080	0.088	0.079	0.080	0.088	0.079
$\hat{\sigma}_\epsilon$	0.019	0.019	0.019	0.019	0.019	0.019	0.019	0.019	0.019
(Intercept)	-0.004	-0.001	-0.005	—	—	—	—	—	—
AGEN	0.039	0.043	0.037	0.002	—	0.002	—	—	—
BANK	0.132	0.125	0.132	0.112	0.104	0.112	0.130	0.114	0.129
DEN	0.136	0.136	0.137	0.112	0.098	0.114	0.115	0.112	0.120
FAM	-0.097	-0.118	-0.092	-0.080	-0.094	-0.076	-0.100	-0.116	-0.099
INEF	-0.125	-0.119	-0.125	-0.103	-0.097	-0.106	-0.131	-0.112	-0.131
RGDP	0.381	0.398	0.375	0.485	0.526	0.473	0.485	0.526	0.473
RIRS	-0.031	-0.030	-0.031	-0.021	-0.016	-0.021	-0.030	—	-0.032
SCHOOL	0.052	0.039	0.055	0.016	—	0.020	—	—	—
TRUST	0.079	0.102	0.073	0.053	0.069	0.049	—	0.080	—
VAAGR	-0.017	-0.019	-0.016	-0.002	-0.003	-0.002	—	—	—
WAGEN	0.021	0.020	0.021	0.002	—	0.003	—	—	—
WBANK	0.013	0.009	0.014	—	—	—	—	—	—
WDEN	0.002	0.003	0.002	—	—	—	—	—	—
WFAM	0.042	0.037	0.043	—	—	0.004	—	—	—
WINEF	0.007	0.009	0.007	—	—	—	—	—	—
WRGDP	0.004	0.013	0.003	0.017	0.018	0.016	—	—	—
WRIRS	-0.033	-0.032	-0.033	-0.007	-0.005	-0.008	—	—	—
WSCHOOL	0.036	0.032	0.037	0.020	0.016	0.021	—	—	—
WTRUST	0.014	0.014	0.014	0.003	0.002	0.004	—	—	—
WVAAGR	0.013	0.015	0.012	—	—	—	—	—	—

Notes: Reported are the results for the generalized method of moments (GMM), L_2 -boosting (LTB) and LTB with post-hoc deselection (DES) for [Anselin \(1988\)](#) random effects model (ANS), [Kapoor et al \(2007\)](#) random effects model (KKP) and the generalized spatial panel model with error components (GSPECM).

The estimated coefficients for all estimation strategies and different model specifications can be seen in Table 5. Regarding the variable selection performance, the results for all considered strategies are comparable. Particularly, LTB reduces the number of variables from 21 to 12 – 16 depending on the particular random effects model. The sparsest results are obtained for the KKP random effects model, while the results for the GSPECM are worst. Further improvement in variable selection can be seen in DES. Particularly, the number of included variables is further reduced to only 6 across all considered random effects models, where spatial lags of independent variables are generally not selected.

Estimates of the spatial autocorrelation parameters $\hat{\rho}_1$ and $\hat{\rho}_2$ are both positive for the GSPECM and KKP random effects model, indicating persistent spatial dependence between disturbances of neighbouring provinces in Italy. Furthermore, estimates of the variances of the disturbances $\hat{\sigma}_\mu$ and $\hat{\sigma}_\epsilon$ remain comparable across all considered random effects models in terms of magnitude.

Moreover, the sign and overall direction of the coefficients are similar across all estimation strategies. The main difference lies in their magnitude, which is affected by the shrinkage induced through early stopping in both LTB and DES. Taking the results

for DES for the GSPECM as an example, the most important variables explaining the non-life insurance in Italian provinces in the years 1998 – 2002 are the the real bank deposits per capita, density of inhabitants per square kilometre, the average number of family members, the judicial inefficiency, the real gross domestic product per capita and the real interest rate on borrowing. Since the independent variables are transformed by simple scaling and centering, the coefficients have an intuitive and simple interpretation. For instance, an increase in average number of family members by one person *ceteris paribus* decreases the real per capita non-life insurance consumption by 0.099 euros on average. Consequently, holding all other variables constant, an increase in the gross domestic product per capita by one unit is associated with an average 0.473 euros increase in real per capita non-life insurance consumption.

4.2 Rice Production in Indonesian Farms

As a second empirical illustration, a spatial panel data set of 171 rice farms in the Cimanuk River Basin, West Java, Indonesia, previously analysed by [Horrace and Schmidt \(1996, 2000\)](#); [Druska and Horrace \(2004\)](#); [Millo and Carmeci \(2011\)](#) and [Millo \(2022\)](#) is considered. The farms were observed over six growing seasons, three wet and three dry, resulting in a balanced panel with $N_{\text{Indonesia}} = 171$ farms and $T_{\text{Indonesia}} = 6$ seasons. Data were collected by the Agro Economic Survey as part of the Rural Dynamic Study and obtained from the Center for Agro Economic Research, Ministry of Agriculture, Indonesia. The data set includes a range of farm-level socio-economic, agronomic, and seasonal environmental variables, which are summarized in Table 6. Most considered variable are time-varying, thereby enabling a fixed effects specification.

Table 6 Data type and description of dependent and independent variables for the rice production in Indonesian farms data set (see, for example, [Horrace and Schmidt \(1996, 2000\)](#); [Druska and Horrace \(2004\)](#); [Millo and Piras \(2012\)](#)).

Variable	Type	Description
GOUTPUT	Numeric	Gross output of rice in kilogram
BIMAS	Categorical	Part of the Bimbingan Masal intensification program (NO/YES/MIXED)
FAMLABOR	Numeric	Family labor in hours
HIREDLABOR	Numeric	Hired labor in hours
PESTICIDE	Numeric	Pesticide cost in Indonesian Rupiah
PHOSPHATE	Numeric	Phosphate in kilogram
PPHOSPH	Numeric	Price of phosphate in Indonesian Rupiah per kilogram
PRICE	Numeric	Price of rough rice in Indonesian Rupiah per kilogram
PSEED	Numeric	Price of seed in Indonesian Rupiah per kilogram
PUREA	Numeric	Price of urea in Indonesian Rupiah per kilogram
SEED	Numeric	Seed in kilogram
SIZE	Numeric	Total area cultivated with rice in hectares
STATUS	Categorical	Land ownership status (OWNER/SHARE/MIXED)
TOTLABOR	Numeric	Total labor hours, excluding harvest labor hours
UREA	Numeric	Urea in kilogram
VARIETIES	Categorical	Varieties of planted rice (TRAD/HIGH/MIXED)
WAGE	Numeric	Labour wage in Indonesian Rupiah per hour

The determinants of rice production on Indonesian farms are investigated using a data-driven variable selection approach based on LTB and DES. Both fixed and random effects specifications of the GSPECM are reported. As a benchmark, GMM estimates for each specification are additionally reported to enable direct comparison. All independent variables listed in Table 6, together with their corresponding spatial lags and an intercept, are included in the estimation. This results in a low-dimensional linear setting with 39 coefficients to be estimated in a non-parsimonious model. Furthermore, continuous variables are log-transformed or scaled via division by 100 to ensure comparability across variables in terms of magnitude and to ensure proper convergence of the LTB algorithm. The implementation of LTB and DES follows closely the Monte Carlo experiments in Section 3. The learning rate is set to $s = 0.1$ and the threshold parameter is fixed at $\tau = 0.01$. The optimal stopping criterion m_{opt} is selected by minimizing the empirical risk using 6-fold leave-time-out cross-validation as discussed by Schratz et al (2024).

Table 7 presents coefficient estimates for rice production in Indonesian farms under both random and fixed effects specifications of the GSPECM. In terms of variable selection, the results of the considered strategies show some divergence. Specifically, under the random effects specification, LTB reduces the number of variables from 39 to 19, whereas under the fixed effects specification, it reduces the number of variables only to 29, indicating considerably weaker performance. In contrast, the improvement of DES for both random and fixed effects specification are comparable, reducing the number of variables to 5 and 6 respectively, including spatial lags of independent variables. Furthermore, the remaining variables are almost identical across both specifications.

Estimates of the spatial autocorrelation parameters $\hat{\rho}_1$ and $\hat{\rho}_2$ are both positive in the random effects specification, indicating strong persistent spatial dependence between neighbouring farms in Indonesia. However, the estimate of $\hat{\rho}_1$ lies at the boundary of the admissible parameter range, casting doubt on the inclusion and estimation of the spatial autocorrelation parameter in the location-specific disturbances.

Moreover, the sign and overall direction of the coefficients remain similar across both specifications. The primary difference lies in the magnitude of the coefficients, which varies due to the shrinkage effects introduced by early stopping in LTB and DES. Taking the results for DES as an example, the most important variables explaining the rice production in Indonesian farms are the seed in kilogram, the total area cultivated with rice in hectares, the total labor hours, urea in kilogram, and the price of phosphate in Indonesian Rupiah per kilogram in neighbouring farms. The interpretation is now based on percentages due to the log-transformation. For instance, an one percent increase in seed usage in kilogram *ceteris paribus* increases rice production by 0.122 percent in kilogram on average. Consequently, holding all other variables constant, an increase in the cultivated area in hectares by one percent is associated with an average 0.509 percent increase in rice production in kilogram.

4.3 Life Expectancy in German Districts

The final empirical illustration is concerned with investigating the drivers of life expectancy in German districts, which is based on the "Indicators and Maps on Spatial and Urban Development in Germany and Europe" (INKAR) database. The INKAR

Table 7 Coefficient estimates across different estimation strategies for the rice production in Indonesian farms data set.

Variable	Random			Fixed		
	GMM	LTB	DES	GMM	LTB	DES
$\hat{\rho}_1$	0.989	0.989	0.989	—	—	—
$\hat{\rho}_2$	0.470	0.470	0.470	0.470	0.470	0.470
$\hat{\sigma}_\mu$	0.012	0.012	0.012	—	—	—
$\hat{\sigma}_\epsilon$	0.073	0.073	0.073	0.073	0.073	0.073
(Intercept)	-5.407	0.331	—	—	—	—
BIMAS MIXED	-0.082	-0.038	—	-0.076	-0.065	—
BIMAS YES	0.086	0.029	—	0.050	0.023	—
FAMLABOR	-0.001	—	—	-0.003	—	—
HIREDLABOR	0.011	0.012	—	0.010	0.017	—
PESTICIDE	0.005	0.003	—	0.003	0.003	—
PHOSPHATE	0.716	0.600	—	0.693	0.768	—
PPHOSPH	-0.772	—	—	-0.364	-0.046	—
PRICE	0.014	—	—	0.041	—	—
PSEED	-0.057	—	—	-0.044	-0.039	—
PUREA	0.726	—	—	0.277	—	—
SEED	0.106	0.109	0.122	0.094	0.092	0.108
SIZE	0.503	0.492	0.509	0.483	0.488	0.499
STATUS MIXED	0.007	0.005	—	0.028	0.039	—
STATUS SHARE	0.155	—	—	0.200	0.153	—
TOTLABOR	0.215	0.224	0.227	0.217	0.202	0.231
UREA	0.131	0.123	0.138	0.103	0.098	0.101
VARIETIES HIGH	0.084	0.044	—	0.066	0.074	—
VARIETIES MIXED	0.117	0.026	—	0.128	0.126	—
WAGE	0.080	—	—	0.136	0.108	—
WBIMAS MIXED	0.083	0.149	—	0.094	0.316	—
WBIMAS YES	1.125	—	—	1.035	0.259	—
WFAMLABOR	-0.178	—	—	-0.118	—	—
WHIREDLABOR	-0.297	-0.013	—	-0.264	-0.118	—
WPESTICIDE	-0.009	—	—	-0.010	-0.014	—
WPHOSPHATE	6.329	3.942	—	6.848	8.977	5.473
WPPHOSPH	-9.808	0.597	1.230	-9.153	—	—
WPRICE	-1.017	—	—	-1.044	-0.570	—
WPSEED	0.669	—	—	0.617	0.148	—
WPUREA	12.717	0.306	—	12.097	1.938	0.901
WSEED	-0.154	—	—	-0.115	—	—
WSIZE	0.062	—	—	0.049	-0.036	—
WSTATUS MIXED	-0.472	0.241	—	-0.510	-0.300	—
WSTATUS SHARE	2.828	—	—	2.714	1.887	—
WTOTLABOR	0.384	—	—	0.274	—	—
WUREA	0.133	—	—	0.144	-0.036	—
WVARIETIES HIGH	-0.688	—	—	-0.633	-0.235	—
WVARIETIES MIXED	0.387	—	—	0.433	0.640	—
WWAGE	-0.174	—	—	-0.210	—	—

Notes: Reported are the results for the generalized method of moments (GMM), L_2 -boosting (LTB) and LTB with post-hoc deselection (DES) for the random effects specification (Random) and fixed effects specification (Fixed) in the generalized spatial panel model with error components (GSPECM).

database contains a variety of socio-demographic, socio-economic, and environmental indicators available on German district level.⁷ The constructed spatial panel data set includes $N_{\text{Germany}} = 400$ districts in which the average life expectancy of a newborn in years has been observed over a time period of $T_{\text{Germany}} = 5$ years, particularly from 2014 – 2019. More than 23 independent variable of interest can be identified, which are summarized in Table 8. Most considered variable are time-invariant or have little variability over time, deeming a fixed effects specification inappropriate (Bundesinstitut für Bau-, Stadt- und Raumforschung (BBSR), 2024).

Table 8 Data type and description of dependent and independent variables for German district data set (see, Balzer (2025)).

Variable	Type	Description
LIFE	Numeric	Average life expectancy of a newborn in years
ACADEMICS	Numeric	Share of socially insured employees at the workplace with an academic professional qualification among socially insured employees in percent
ACC	Numeric	Total road traffic accidents per 100,000 inhabitants
AGE	Numeric	Average age of the population in years
CAR	Numeric	Passenger cars per 1,000 inhabitants
COM	Numeric	Commuter balance per 100 socially insured employees at the workplace
DEBT	Numeric	Percentage of private debtors among residents aged 18 and over
DIVORCES	Numeric	Divorces per 1,000 inhabitants aged 18 and over
EMPLOYMENT	Numeric	Percentage of social insurance contributing employees per 100 working-age residents at place of residence
FOREIGN	Numeric	Share of foreigners among the residents in percent
GDP	Numeric	Gross domestic product per inhabitant
HHINC	Numeric	Average household income per inhabitant in euro
INS	Numeric	Consumer insolvency proceedings per 1,000 inhabitants aged 18 and over
LABOR	Numeric	Number of hours worked by employees
LAND	Numeric	Cadastral area in square kilometers
LIVE	Numeric	Living space per inhabitant in square meters
MEDINC	Numeric	Median income of full-time employees subject to social security contributions in euro
PART	Numeric	Labor force participation rate among the working-age population in percent
POP	Numeric	Inhabitants per square kilometer
RENT	Numeric	Rent prices per square meter in euro
SELF	Numeric	Percentage of self-employed among all employed individuals
TAX	Numeric	Tax revenue per inhabitant in euro
UNEMPLOYMENT	Numeric	Percentage of unemployed among the civilian labor force
WELFARE	Numeric	Share of employable and non-employable welfare recipients among residents under 65 years old

Thus, to investigate the drivers of life expectancy in German districts from a data-driven variable selection perspective via LTB and DES, a variety of random effects

⁷The INKAR database in the current as well as the 2021 version with the codebook for all indicators is freely available at <https://www.inkar.de/>

models, namely ANS, KKP, and GSPECM, are considered. For additional comparison, results from classical GMM estimation are also reported. All independent variables presented in Table 8, their corresponding spatial lags, as well as an intercept, are included in the estimation, leading to a low-dimensional linear setting with 47 coefficients to estimate in a non-parsimonious model. Furthermore, continuous variables are centered and scaled to ensure comparability across variables and convergence of the LTB algorithm. The setup for LTB and DES closely follows the Monte Carlo experiments in Section 3, with the learning rate set to $s = 0.1$ and the threshold to $\tau = 0.01$. The spatial weight matrix is constructed using a 10-nearest-neighbor structure, such that each district is assigned its ten geographically closest neighbors and is subsequently row-normalized. The optimal stopping criterion m_{opt} is determined by minimizing the empirical risk via 10-fold spatial cross-validation as discussed by Brenning (2012).⁸

The estimated coefficients for all estimation strategies and different model specifications can be seen in Table 9. For brevity, the coefficients of the spatial lags of exogenous variables are not reported individually. Instead, the number of spatial lags of variables included in each model (No. $\mathbf{W}\mathbf{X}$) is presented. Regarding the variable selection performance, the results for all considered strategies are comparable. Particularly, LTB reduces the number of variables from 47 to 36 – 38 depending on the particular random effects model. The sparsest results are obtained for the ANS random effects model, while the results for the GSPECM are worst, although the range is negligible. Further improvement in variable selection can be seen in DES. Particularly, the number of included variables is further reduced to only 7 – 8 across all considered random effects models, where spatial lags of independent variables are not selected.

Estimates of the spatial autocorrelation parameters are negative for $\hat{\rho}_1$ and positive for $\hat{\rho}_2$ in the GSPECM, indicating diverging results regarding the persistent spatial dependence between neighboring districts in Germany. In contrast, estimates of the variances of the disturbances $\hat{\sigma}_{\mu}$ and $\hat{\sigma}_{\epsilon}$ remain comparable across all considered random effects models.

Moreover, the algebraic sign and overall direction of the coefficients remain identical across all estimation strategies. The primary difference lies in the magnitude of the coefficients, which varies due to the shrinkage effects introduced by early stopping in LTB and DES. Taking the results for DES for the GSPECM as an example, the most important variables explaining the life expectancy in German districts in the years 2014 – 2019 are the proportion of employees with an academic degree, the average age of the population, the debt quota, the household income, the rent prices, the unemployment rate and the share of welfare recipients. Since the independent variables are transformed by simple scaling and centering, the coefficients have an intuitive and simple interpretation. For instance, an increase in rent prices by one euro ceteris paribus increases the average life expectancy by 0.100 years on average. Consequently, holding all other variables constant, an one percentage point increase in the share of private debtors is associated with an average 0.368 year decrease in average life expectancy.

To summarize the findings, the spatial autocorrelation parameters reveal diverging spatial dependence between districts, that is, the disturbances in one district are

⁸The shapefile of the German districts is provided freely by the Federal Agency for Cartography and Geodesy and can be found at https://gdz.bkg.bund.de/index.php/default/digitale-geodaten/verwaltungsgebiete.html?__store=default.

influenced by the disturbances in neighboring districts in the opposite direction. Furthermore, the share of welfare recipients and average age of the population negatively influence average life expectancy meaning that underlying poor economic and health status decrease the average life expectancy. Furthermore, districts with higher unemployment and debt quotas have on average lower average life expectancies, indicating that financial hardship, fiscal stress or economic precarity is negatively associated with longevity. In contrast, higher rent prices, higher household income and higher share of employees with an academic degree lead to a higher average life expectancy, which shows that inhabitants in wealthier districts and with better socioeconomic status live longer on average (Bundesinstitut für Bau-, Stadt- und Raumforschung (BBSR), 2024; Bundesamt für Kartographie und Geodäsie (BKG), 2024; Balzer, 2025).

Table 9 Coefficient estimates across different estimation strategies for the German life expectancy data set.

	GMM			LTB			DES		
	ANS	KKP	GSPECM	ANS	KKP	GSPECM	ANS	KKP	GSPECM
$\hat{\rho}_1$	0.000	0.212	-0.284	0.000	0.212	-0.284	0.000	0.212	-0.284
$\hat{\rho}_2$	0.212	0.212	0.212	0.212	0.212	0.212	0.212	0.212	0.212
$\hat{\sigma}_\mu$	0.148	0.146	0.144	0.148	0.146	0.144	0.148	0.146	0.144
$\hat{\sigma}_\epsilon$	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050
(Intercept)	0.001	0.002	0.002	—	—	—	-0.002	—	0.002
ACADEMICS	0.284	0.284	0.283	0.250	0.250	0.250	0.179	0.181	0.184
ACC	0.029	0.028	0.028	0.021	0.021	0.021	—	—	—
AGE	-0.096	-0.094	-0.092	-0.096	-0.096	-0.097	-0.110	-0.111	-0.114
CAR	0.057	0.056	0.057	0.037	0.036	0.036	—	—	—
COM	-0.048	-0.048	-0.048	-0.049	-0.042	-0.042	—	—	—
DEBT	-0.334	-0.334	-0.335	-0.356	-0.351	-0.347	-0.380	-0.375	-0.368
DIVORCES	0.024	0.024	0.025	0.023	0.023	0.023	—	—	—
EMPLOYMENT	-0.055	-0.060	-0.065	-0.020	-0.025	-0.032	—	—	—
FOREIGN	0.082	0.081	0.080	0.061	0.059	0.056	—	—	—
GDP	-0.101	-0.102	-0.101	-0.092	-0.093	-0.091	—	—	—
HHINC	0.211	0.211	0.210	0.207	0.206	0.205	0.247	0.246	0.246
INS	-0.010	-0.010	-0.009	-0.014	-0.014	-0.014	—	—	—
LABOR	-0.073	-0.075	-0.077	-0.071	-0.073	-0.076	—	—	—
LAND	-0.008	-0.008	-0.008	-0.007	-0.005	-0.004	—	—	—
LIVE	-0.108	-0.111	-0.114	-0.075	-0.078	-0.083	—	—	—
MEDINC	0.070	0.069	0.067	0.062	0.065	0.065	0.009	0.006	—
PART	0.084	0.085	0.086	0.057	0.060	0.063	—	—	—
POP	-0.081	-0.083	-0.083	-0.051	-0.054	-0.059	—	—	—
RENT	0.081	0.080	0.079	0.099	0.099	0.097	0.098	0.099	0.100
SELF	0.056	0.058	0.060	0.043	0.048	0.051	—	—	—
TAX	-0.037	-0.037	-0.036	-0.028	-0.027	-0.027	—	—	—
UNEMPLOYMENT	-0.076	-0.079	-0.084	-0.088	-0.087	-0.087	-0.048	-0.051	-0.055
WELFARE	-0.093	-0.090	-0.087	-0.060	-0.063	-0.066	-0.112	-0.109	-0.107
No. $\mathbf{WX}$	23	23	23	13	14	15	0	0	0

Notes: Reported are the results for the generalized method of moments (GMM), L_2 -boosting (LTB) and LTB with post-hoc deselection (DES) for Anselin (1988) random effects model (ANS), Kapoor et al (2007) random effects model (KKP) and the generalized spatial panel model with error components (GSPECM). For brevity, the coefficients of the spatial lags of exogenous variables are not reported individually. Instead, the number of spatial lag variables included in each strategy (No. $\mathbf{WX}$) is presented.

5 Conclusion

The key findings and main contributions of this article are: (a) Model-based gradient boosting is extended for the GSPECM, which nests important spatial cases such as the ANS and KKP random effects model. (b) Instead of relying on a spatial error family in practice, a Cochrane–Orcutt type spatial transformation on the data is used, such that model-based gradient boosting reduces to LTB with the L_2 loss function. The major advantage of such an approach is the preservation of theoretical properties such as convergence and consistency, which cannot be ensured via the spatial error family. (c) The Cochrane–Orcutt type spatial transformation relies on the knowledge about the spatial autocorrelation parameters ρ_1 and ρ_2 , which are typically unknown in practice, making LTB infeasible. Thus, feasible LTB relies on replacing unknown quantities by estimates based on estimators proposed by [Kapoor et al \(2007\)](#) and [Baltagi and Liu \(2016\)](#). (d) Monte Carlo experiments confirm proper functionality of LTB in low- and high-dimensional settings with respect to variable selection and estimation accuracy across varying combinations of spatial autocorrelation parameters and outperform traditional ML and GMM estimators. (e) The application of LTB is illustrated by revisiting classical spatial econometric panel data sets, namely non-life Insurance in Italian provinces, rice production in Indonesian farms and life expectancy in German districts.

Additionally, limitations, improvements and extensions beyond the scope of this article have to be acknowledged. Although Monte Carlo experiments have been conducted, the scope of the experiments is by no means exhaustive. More complex scenarios such as varying spatial weight matrices or varying data generating process can be considered, which might give a more differentiated view of LTB for the GSPECM. The current implementation of LTB for the GSPECM does not include a spatial lag of the dependent variable. Thus, extending LTB for mixed autoregressive spatial panel models with error components is an interesting avenue for further research. To mitigate consequences arising from low TNR values such as the inclusion of too many non-informative variables, impacting the sparsity of the final model, post-hoc deselection is utilized. However, alternative approaches such as stability selection might be applied to control over the amount of false positive variables ([Meinshausen and Bühlmann, 2010](#); [Shah and Samworth, 2012](#); [Hofner et al, 2015](#)). Since the Cochrane–Orcutt type spatial transformed GSEPCM is a univariate location model, sparser models can be obtained by applying probing, which stops LTB as soon as the first randomly permuted version of a variable is added ([Thomas et al, 2017](#)). Additionally, p-values for individual base-learners can be computed by utilizing permutation techniques ([Hepp et al, 2019](#)).

Therefore, the next goal is to extend LTB for mixed autoregressive spatial panel models with error components, which is especially challenging in high-dimensional settings due to the increasing model complexity. However, such an extension allows for a more nuanced examination of the empirical illustrations. Furthermore, the development of an efficient software implementation such as a dedicated package in the programming language R is an important avenue for further research. Nevertheless, practitioners and applied statisticians working with spatial panel data are encouraged to utilize LTB for the GSPECM and various nested models such as the ANS and KKP

random effects model as a valuable alternative for estimation, regularization, model and variable selection in future research.

Supplementary information. All R-code for the implemented L_2 -boosting algorithm along with the Monte Carlo experiments is publicly available in the following GitHub repository <https://github.com/micbalz/SpatPanelRegBoost>.

Declarations

Data availability. The data as well as the accompanying codebook on the life expectancy in German districts is publicly available via <https://www.inkar.de/> (DL-DE BY 2.0). The data and the codebook for the non-life insurance in Italian provinces and rice production in Indonesian farms is publicly available in the `splm` package in R (Millo and Piras, 2012; Bivand et al, 2021). The shapefile for the North Carolina counties is available in the `spData` package in R (Pebesma and Bivand, 2023; Bivand et al, 2025).

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